

BAD702: SMD Module 1

Question 1:

Explain the different measures of central tendency and variability used in EDA.
Provide suitable examples.

Answer:

1. Introduction

Exploratory Data Analysis (EDA) uses statistical measures to understand the **central location, spread and distribution** of data.

The two important groups of measures are:

1. **Measures of Central Tendency** – describe the typical or central value of the data.
 2. **Measures of Variability** – describe how much the data values differ or spread around the central value.
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2. Measures of Central Tendency

Central tendency represents the **central or typical value** of a dataset.

The main measures are:

A. Mean

The **arithmetic mean** is obtained by dividing the sum of all observations by the number of observations.

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

where:

- x_i = individual observation
- n = number of observations.

Example

Consider:

2, 4, 6, 8, 10

$$\bar{x} = \frac{2 + 4 + 6 + 8 + 10}{5} = \frac{30}{5} = \boxed{6}$$

Advantage: Simple and widely used.

Limitation: It is strongly affected by **outliers**.

B. Median

The **median** is the middle value when observations are arranged in ascending or descending order.

For an odd number of observations:

$$\text{Median} = x_{\frac{n+1}{2}}$$

For an even number of observations:

$$\text{Median} = \frac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2}$$

Example

For:

2, 4, 6, 8, 10

the middle value is:

Median = 6

The median is more **robust to outliers** than the mean.

C. Mode

The **mode** is the value that occurs **most frequently** in the dataset.

Example

2, 3, 3, 4, 5, 3, 6

Here, 3 occurs most frequently.

Mode = 3

Mode is particularly useful for **categorical data**.

3. Measures of Variability

Variability measures the **spread or dispersion** of observations around the central value.

Important measures include:

A. Range

The **range** is the difference between the maximum and minimum values.

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

For:

2, 4, 6, 8, 10

$$\text{Range} = 10 - 2 = \boxed{8}$$

Limitation: It depends only on the two extreme observations and is therefore sensitive to outliers.

B. Variance

Variance measures the average squared deviation of observations from the mean.

For a population:

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2$$

For a sample:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

where:

- μ = population mean
- $\bar{x}$ = sample mean.

A larger variance indicates **greater spread**.

C. Standard Deviation

Standard deviation is the square root of variance.

$$\sigma = \sqrt{\sigma^2}$$

For the data:

2, 4, 6, 8, 10

the mean is 6.

Squared deviations are:

$$(2 - 6)^2 = 16$$

$$(4 - 6)^2 = 4$$

$$(6 - 6)^2 = 0$$

$$(8 - 6)^2 = 4$$

$$(10 - 6)^2 = 16$$

Population variance:

$$\sigma^2 = \frac{16 + 4 + 0 + 4 + 16}{5} = \frac{40}{5} = 8$$

Therefore:

$$\sigma = \sqrt{8} \approx 2.83$$

Standard deviation is expressed in the **same units as the original data**, making it easier to interpret than variance.

D. Interquartile Range (IQR)

The **Interquartile Range (IQR)** measures the spread of the middle 50% of observations.

$$IQR = Q_3 - Q_1$$

where:

- Q_1 = first quartile
- Q_3 = third quartile.

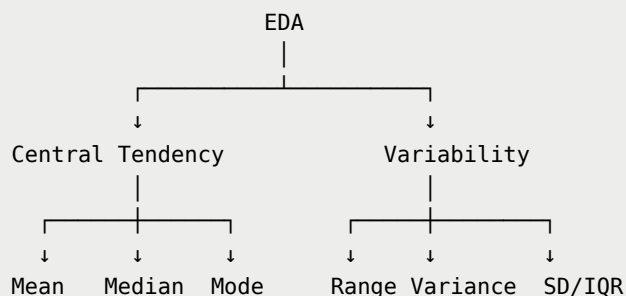
IQR is **less affected by outliers** and is therefore useful for skewed data.

4. Summary

Measure	Purpose	Main Characteristic
Mean	Central value	Sensitive to outliers
Median	Central value	Robust to outliers
Mode	Most frequent value	Useful for categorical data
Range	Total spread	Depends on extremes
Variance	Average squared spread	Units are squared
Standard Deviation	Spread around mean	Same units as data
IQR	Spread of middle 50%	Robust to outliers

5. Importance in EDA

These measures help an analyst quickly understand the dataset:





Question 2:

Compare and contrast univariate, bivariate and multivariate data in Exploratory Data Analysis (EDA).

Answer

1. Introduction

In **Exploratory Data Analysis (EDA)**, data can be analyzed according to the **number of variables being studied simultaneously**.

The three major types are:

1. **Univariate analysis** – analysis of one variable.
2. **Bivariate analysis** – analysis of two variables.
3. **Multivariate analysis** – analysis of more than two variables.

The choice of statistical summaries and visualizations depends on whether the variables are **numeric or categorical**.

2. Univariate Analysis

Univariate analysis involves studying **one variable at a time**. Its main objective is to understand the **distribution, central tendency and variability** of that variable.

Common measures

- **Mean**
- **Median**
- **Percentiles**

- **Variance**
- **Standard deviation**
- **Range**
- **Interquartile range (IQR)**

Common visualizations

For a **numeric variable**:

- **Histogram**
- **Boxplot**
- **Density plot**

For a **categorical variable**:

- **Frequency table**
- **Bar chart**

A histogram shows frequency counts, while a boxplot provides a quick view of the distribution and potential outliers.

Example

If we study only **student marks**:

60, 65, 70, 75, 80

we can calculate their mean, median, standard deviation and visualize them using a histogram or boxplot.

3. Bivariate Analysis

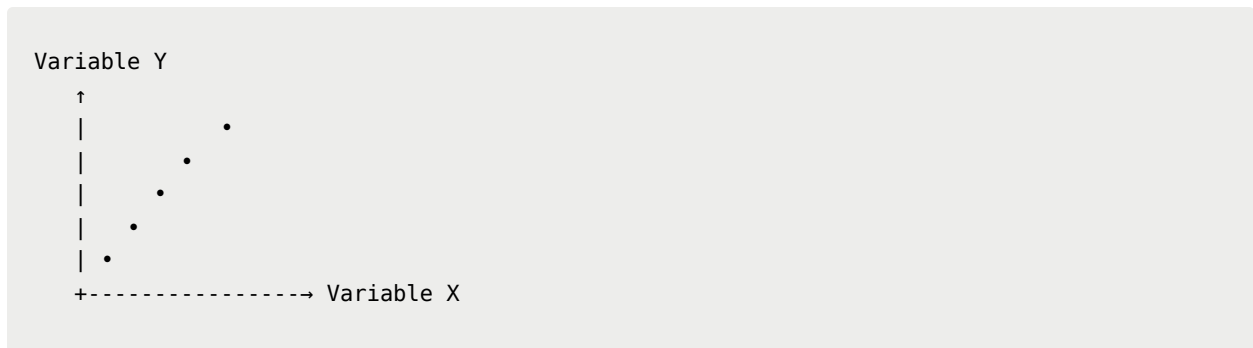
Bivariate analysis involves studying the **relationship between two variables**.

Correlation analysis is an important method of bivariate analysis.

The analysis depends on the type of the two variables.

A. Numeric vs Numeric

A **scatterplot** is the standard visualization for studying the relationship between two measured numeric variables.



Other techniques include:

- **Hexagonal binning**
- **Contour plots**
- **Heat maps**

These are useful when a scatterplot becomes too dense for very large datasets.

Example

Study the relationship between:

- **House size**
- **House tax-assessed value**

A scatterplot can show whether larger houses tend to have higher assessed values.

B. Categorical vs Categorical

A **contingency table** summarizes counts between two or more categorical variables.

Example:

Loan Grade	Fully Paid	Late	Charged Off
A	80	5	2
B	70	10	5

Loan Grade	Fully Paid	Late	Charged Off
C	55	15	10

This helps identify relationships between **loan grade** and **loan outcome**.

4. Multivariate Analysis

Multivariate analysis involves examining **more than two variables simultaneously**. It helps identify relationships and interactions that may not be visible when variables are examined individually or in pairs.

Common approaches

- **Scatterplot matrix**
- **Heat maps**
- **Multivariate tables**
- **Grouped/conditioned plots**
- **Dimensionality-reduction visualizations**, where appropriate

Example

Suppose a dataset contains:

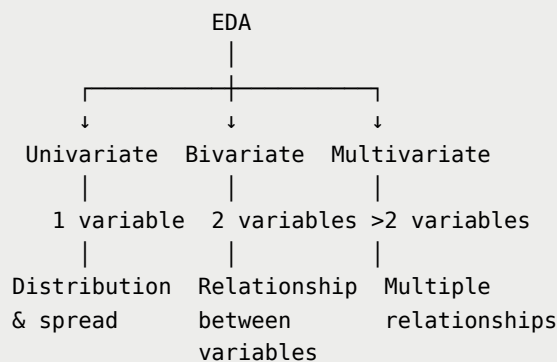
- **House size**
- **Number of bedrooms**
- **Location**
- **Age of house**
- **Tax-assessed value**

Multivariate analysis can examine how several of these variables together relate to **tax-assessed value**.

5. Comparison

Feature	Univariate	Bivariate	Multivariate
Number of variables	1	2	More than 2
Main objective	Understand distribution of one variable	Study relationship between two variables	Study relationships among multiple variables
Statistics	Mean, median, variance, percentiles	Correlation, cross-tabulation	Multiple relationships/interactions
Numeric data visualization	Histogram, boxplot, density plot	Scatterplot, hexagonal binning, contour plot	Scatterplot matrix, heat maps, multivariate plots
Categorical data	Frequency table/bar chart	Contingency table	Multivariate contingency/grouped analysis
Example	Distribution of student marks	Marks vs study hours	Marks vs study hours, attendance and previous GPA

6. Relationship Between the Three



As the number of variables increases, the analysis becomes more complex because the analyst must consider **interactions and relationships among several variables**.

7. Key Distinction

The essential difference is the **number of variables analyzed together**:

Univariate=1

Bivariate=2

Multivariate>2

For example:

- **Univariate:** What is the distribution of house prices?
 - **Bivariate:** How does house size relate to house price?
 - **Multivariate:** How do house size, location, number of bedrooms and age jointly relate to house price?
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Question 3:

Explain how to explore categorical data. Include both summary statistics and visuals.

Answer

1. Introduction

Categorical data represents data divided into distinct categories or levels, such as **gender, product type, loan status, or cause of delay**.

In **Exploratory Data Analysis (EDA)**, categorical data is mainly explored using **proportions or percentages** and visualized using **bar charts** or **pie charts**.

2. Summary Statistics for Categorical Data

A. Frequency / Count

The first step is to determine the **number of observations belonging to each category**.

For example, consider the cause of flight delays:

Cause	Number of Delays
Carrier	23
ATC	30
Weather	4
Security	1
Inbound	42

The frequency tells us how often each category occurs.

B. Proportion or Percentage

For categorical data, **proportions or percentages** provide a useful summary.

$$\text{Proportion of category} = \frac{\text{Frequency of category}}{\text{Total frequency}}$$

Percentage is:

$$\text{Percentage} = \text{Proportion} \times 100$$

For example, if 42 out of 100 delays are caused by **Inbound** aircraft:

$$\text{Percentage} = \frac{42}{100} \times 100 = \boxed{42\%}$$

The notes give an example of Dallas/Fort Worth airport delays:

Cause	Percentage
Carrier	23.02%
ATC	30.40%
Weather	4.03%
Security	0.12%
Inbound	42.43%

Thus, **Inbound** is the most common cause in this example.

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C. Mode

The **mode** is the **most commonly occurring category or value** in a dataset.

For example:

`{Carrier, ATC, Inbound, Inbound, Weather}`

The mode is:

`Inbound`

Mode is a particularly useful summary statistic for **categorical data**.

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D. Expected Value

When categories can be associated with **numeric values**, an **expected value** can summarize the data.

It is calculated as:

$$EV = \sum_i p_i x_i$$

where:

- x_i = value associated with category i
- p_i = probability/proportion of that category.

For example, if a service has values of \$300 and \$50 with probabilities 0.05 and 0.15, while 0.80 purchase nothing:

$$EV = (0.05)(300) + (0.15)(50) + (0.80)(0)$$

$$EV = \$22.50$$

Expected value is essentially a **weighted mean**, with probabilities acting as weights.

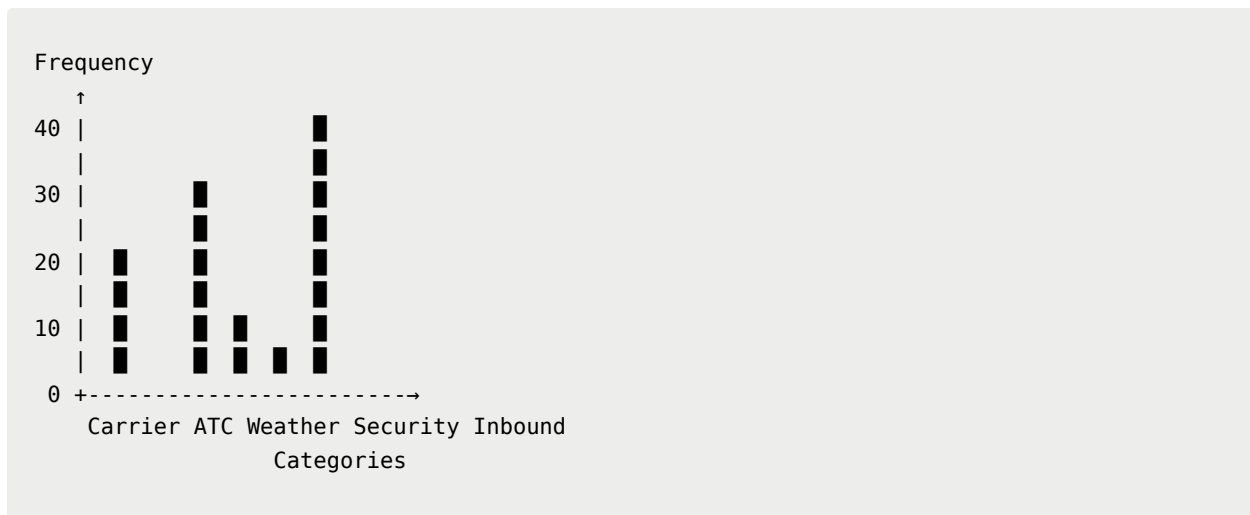
3. Visual Exploration of Categorical Data

A. Bar Chart

A **bar chart** is the most common visual tool for displaying a single categorical variable.

- **X-axis:** categories
- **Y-axis:** frequency or proportion
- Bars are **separate** from each other because categories are distinct.

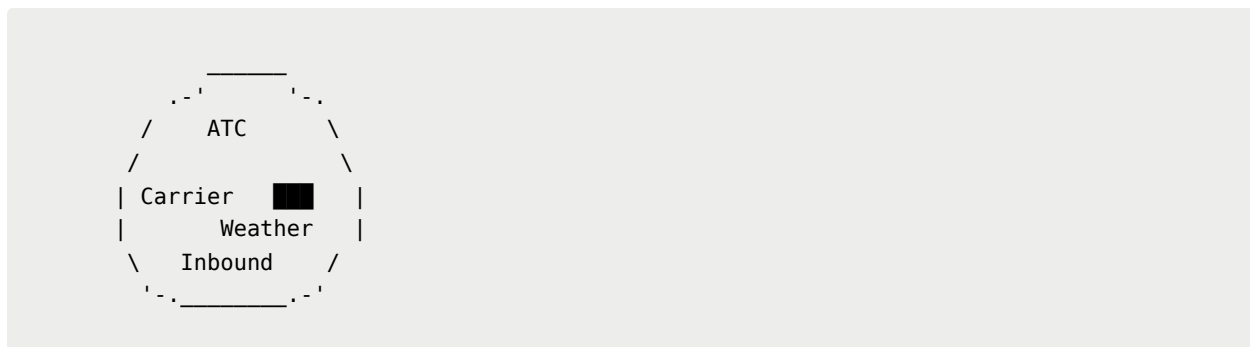
Exam Diagram



The height of each bar represents the **frequency or proportion** of that category.

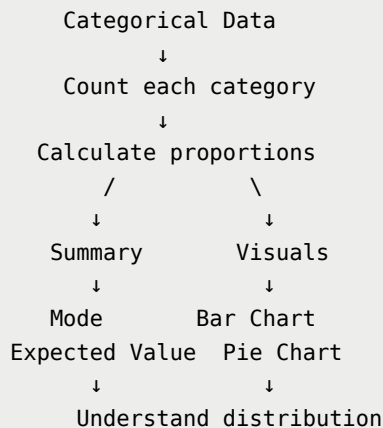
B. Pie Chart

A **pie chart** represents each category as a **wedge of a circle**, where the size of the wedge corresponds to its frequency or proportion.



Pie charts are an alternative to bar charts. However, the notes point out that **statisticians and data-visualization experts generally prefer bar charts**, because pie charts are less visually informative.

4. Categorical Data Exploration Process



5. Important Points

- Categorical data is generally summarized using **counts, proportions and percentages**.
- **Mode** identifies the most frequently occurring category.
- **Expected value** can be used when categories correspond to numerical values.
- **Bar charts** are the primary visual representation.
- **Pie charts** can also be used, but are generally less informative than bar charts.
- Categories can also represent **binned numerical data**, converting continuous numerical values into ordered categories.

QUESTION 4:

Explain skewness and kurtosis in data distribution.

Answer

1. Introduction

Skewness and **kurtosis** are statistical moments used to describe important characteristics of a **data distribution**.

- **Skewness** is the **third statistical moment** and describes whether data is skewed toward **larger or smaller values**.
 - **Kurtosis** is the **fourth statistical moment** and indicates the tendency of the data to contain **extreme values**.
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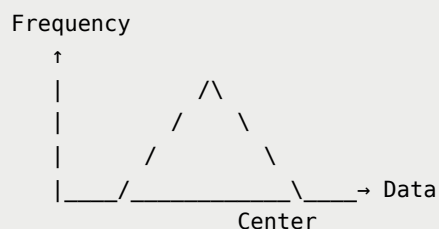
2. Skewness

Skewness describes the **asymmetry of a distribution**. It indicates whether the distribution has a longer tail toward smaller or larger values.

Types of Skewness

A. Symmetric Distribution

A symmetric distribution has approximately equal spread on both sides of its center.

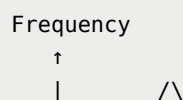


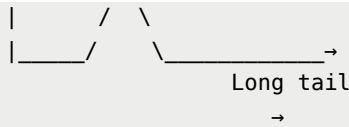
There is no significant skew.

$$\text{Skewness} \approx 0$$

B. Positive Skewness

In a **positively skewed** distribution, the tail extends toward **larger values**.



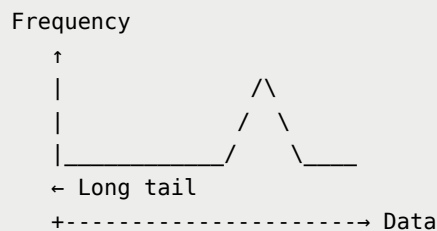


The distribution is therefore **right-skewed**.

Example: Income distribution, where a small number of people may have very high incomes.

C. Negative Skewness

In a **negatively skewed** distribution, the tail extends toward **smaller values**.



The distribution is therefore **left-skewed**.

Key point

Skewness tells the direction of asymmetry

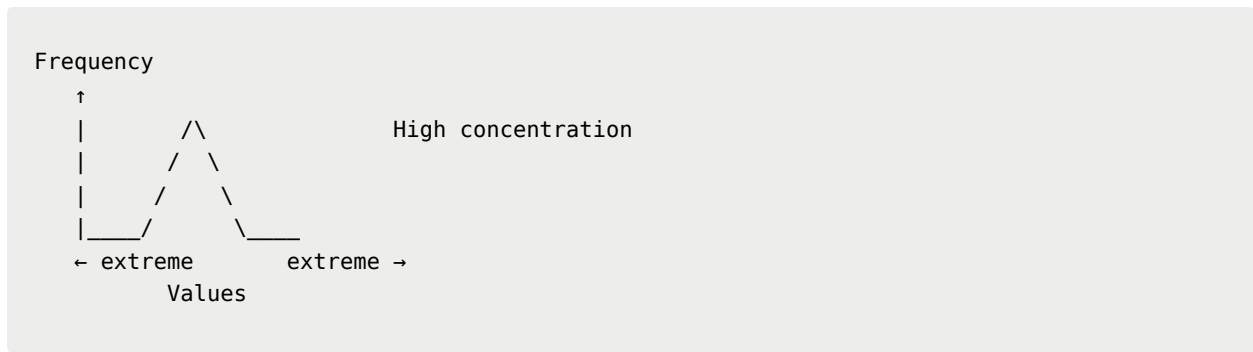
It helps identify whether observations tend to have a longer tail toward **higher or lower values**.

3. Kurtosis

Kurtosis is the **fourth statistical moment**. It indicates the **propensity of the data to have extreme values**.

Thus, kurtosis is useful for understanding the **tail behavior** of a distribution and the presence of extreme observations.

Illustration



A distribution with greater tendency to produce **extreme values** has greater tail heaviness.

4. Statistical Moment Interpretation

The moments provide a useful way to characterize a distribution:

Moment	Statistical Property
First moment	Location / central tendency
Second moment	Variability
Third moment	Skewness
Fourth moment	Kurtosis

Therefore:

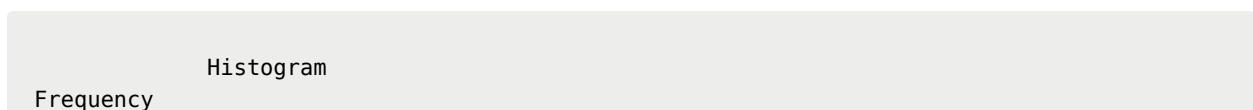
3rd Moment → Skewness

4th Moment → Kurtosis

The notes emphasize that, in practical EDA, skewness and kurtosis are generally **identified through visual displays** rather than relying primarily on numerical metrics. Histograms and density plots are useful for examining the overall distribution.

5. Using EDA Visuals

A **histogram** is particularly useful for observing skewness and extreme values.





A **density plot** can also be used. It represents the distribution as a **continuous, smoothed line** and can be thought of as a smoothed histogram.

6. Importance in EDA

Skewness and kurtosis help an analyst:

- Understand the **shape of a distribution**.
 - Identify whether data is concentrated more toward one side.
 - Detect a tendency toward **extreme values**.
 - Decide whether transformations or robust statistical measures may be appropriate.
 - Better understand the distribution before applying statistical or machine-learning models.
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Question 5:

Explain the different methods used to explore numerical data distributions, including histograms, density plots and boxplots. Discuss their interpretation.

Answer

1. Introduction

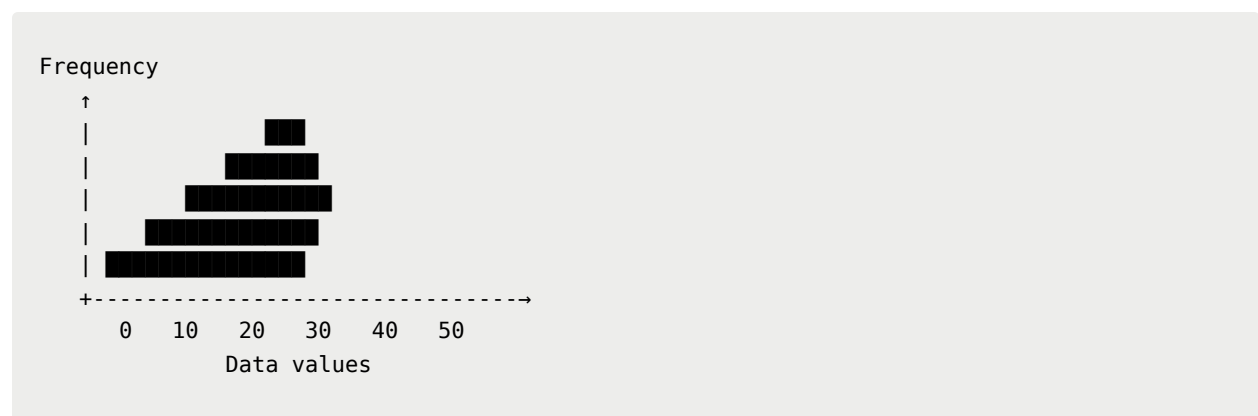
In **Exploratory Data Analysis (EDA)**, numerical data is explored not only through summary statistics but also through **visualizations of the overall distribution**. The important methods are **histograms, density plots and boxplots**.

These plots help identify the **center, spread, shape, skewness, concentration and potential outliers** in numerical data.

2. Histogram

A **histogram** is a graphical representation of a **frequency table**. The numerical values are divided into intervals called **bins**, and the frequency or proportion of observations in each bin is displayed.

Diagram



Characteristics

- **X-axis:** numerical values/bins.
- **Y-axis:** frequency or proportion.
- Bins are generally of **equal width**.
- Bars are **contiguous**; there is normally no gap between adjacent bins.
- Empty bins should also be shown because they provide useful information.

Interpretation

A histogram helps identify:

- **Center** of the distribution.
- **Spread** of the data.
- **Shape** of the distribution.
- **Skewness** toward larger or smaller values.
- **Unusual or extreme observations.**
- Possible **multiple peaks** or clusters.

Important point: Bin size

The choice of bin size is important:

- **Too large:** important features of the distribution may be hidden.
- **Too small:** the graph becomes too granular and the overall pattern becomes difficult to see.

3. Density Plot

A **density plot** is a **smoothed version of a histogram**. It represents the distribution of numerical values as a **continuous line**, generally calculated using a **kernel density estimate**.

Diagram



Characteristics

- The distribution is represented by a **smooth continuous curve**.

- It does not depend directly on discrete histogram bins.
- It is often based on **kernel density estimation (KDE)**.
- It can be superimposed on a histogram to provide a smoother view of the same distribution.

Interpretation

A density plot helps identify:

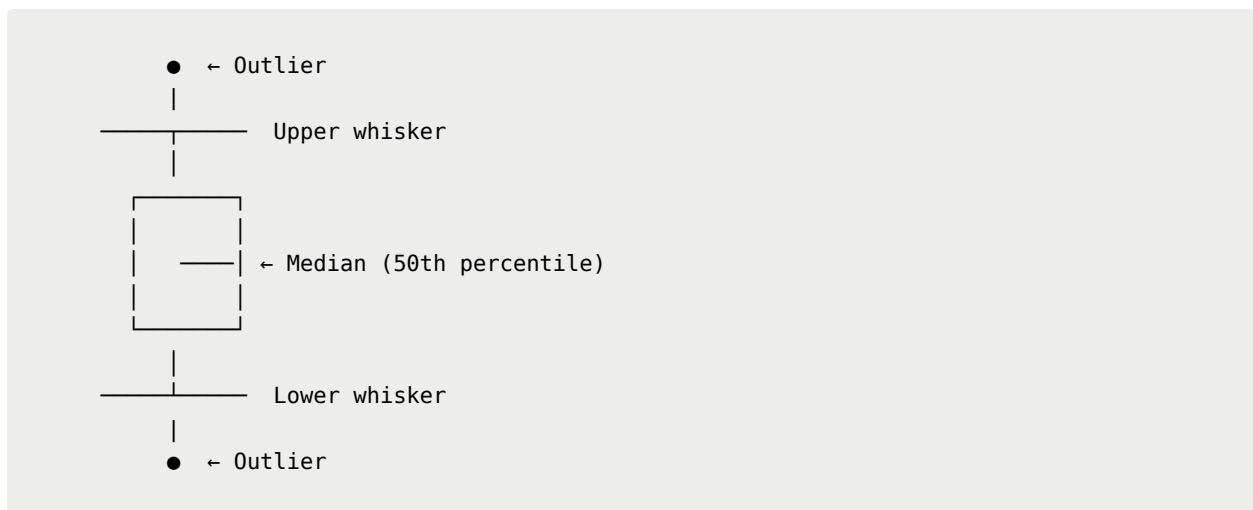
- **Peaks** and areas of high concentration.
- **Skewness.**
- **Spread.**
- **Multiple modes or clusters.**
- The overall shape of the distribution.

Thus, a density plot gives a smoother representation of the distribution than a histogram.

4. Boxplot

A **boxplot**, also called a **box-and-whiskers plot**, provides a quick visual summary of the distribution using **percentiles**.

Diagram



Main Components

- **Lower quartile Q1Q_1**: 25th percentile.
- **Median Q2Q_2**: 50th percentile.
- **Upper quartile Q3Q_3**: 75th percentile.
- **Interquartile range:**

$$\text{IQR} = \text{Q3} - \text{Q1}$$

- **Whiskers**: extend toward the lower and upper parts of the data.
- Individual points beyond the whiskers can indicate **outliers**.

Interpretation

A boxplot quickly shows:

1. **Median** → central location.
2. **Box width/height** → spread of the middle 50% of observations.
3. **Whiskers** → extent of the distribution.
4. **Outliers** → unusually distant observations.
5. **Asymmetry** → can indicate skewness.

Boxplots are particularly useful when **comparing distributions across different groups** because several boxplots can be displayed side by side.

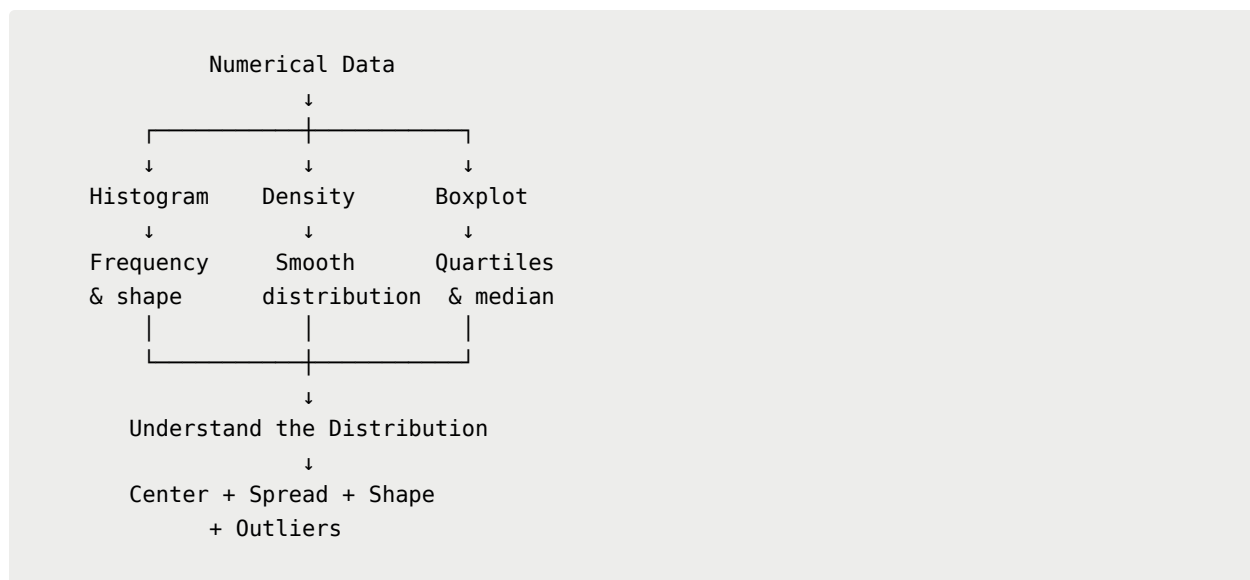
5. Comparison of the Three Methods

Feature	Histogram	Density Plot	Boxplot
Representation	Bars	Smooth curve	Box and whiskers
Main purpose	Show frequency distribution	Show smoothed distribution	Summarize distribution using percentiles
X/Y information	Bins vs frequency	Values vs density	Values vs percentile-based summary
Shows shape	Very well	Very well	Limited

Feature	Histogram	Density Plot	Boxplot
Shows skewness	Yes	Yes	Yes, approximately
Shows outliers	Possible	Less explicit	Very clearly
Shows median	Not directly	Not directly	Yes
Shows quartiles	No	No	Yes
Useful for group comparison	Less convenient	Possible	Very useful

6. Overall Interpretation in EDA

The three methods complement each other:



For example, if a dataset contains **house prices**:

- A **histogram** shows how frequently different price ranges occur.
- A **density plot** shows the smooth shape and concentration of prices.
- A **boxplot** shows the median, quartiles, spread and possible unusually expensive houses.

Question 6:

Explain the role of percentiles, quartiles and interquartile range (IQR) in exploratory data analysis. Discuss their usefulness in identifying the spread and unusual observations in data.

Answer

1. Introduction

Percentiles, quartiles and Interquartile Range (IQR) are important **order statistics** used in Exploratory Data Analysis (EDA). They summarize the distribution of data based on its **sorted values** and provide robust measures of spread that are less affected by extreme observations.

2. Percentiles

A **percentile** is a value such that approximately **P percent of the observations are at or below it**, while the remaining $(100-P)$ percent are at or above it.

For example:

- **25th percentile** → 25% of values are at or below this value.
- **50th percentile** → 50% of values are at or below this value.
- **75th percentile** → 75% of values are at or below this value.
- **95th percentile** → 95% of values are at or below this value.

The **50th percentile is the median**.

Example

Consider:

1, 2, 3, 3, 5, 6, 7, 9

The notes give:

$P_{25}=2.5$

$P_{75}=6.5$

Thus, percentiles allow us to understand where observations lie within the **overall distribution**.

Importance of Percentiles

Percentiles are useful for:

- Summarizing the **entire distribution**.
- Understanding the **tails** of the distribution.
- Comparing an observation with the rest of the data.
- Measuring variability without being strongly affected by extreme values.

For example, reporting the **5th, 25th, 50th, 75th and 95th percentiles** provides a useful description of the distribution.

3. Quartiles

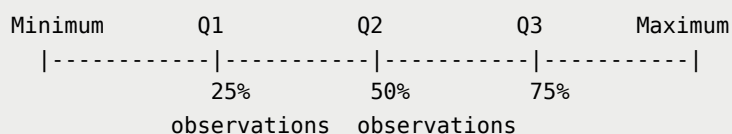
Quartiles are specific percentiles that divide ordered data into approximately four parts.

$Q1 = P25$

$Q2 = P50 = \text{Median}$

$Q3 = P75$

Therefore:



- **Q1Q_1**: first quartile, or 25th percentile.
- **Q2Q_2**: second quartile, or 50th percentile.
- **Q3Q_3**: third quartile, or 75th percentile.

Quartiles are commonly reported to summarize the **location and spread** of a dataset.

4. Interquartile Range (IQR)

The **Interquartile Range (IQR)** measures the spread of the **middle 50% of the data**.

$$\text{IQR} = Q3 - Q1$$

Since $Q1 = P25$ and $Q3 = P75$:

$$\text{IQR} = P75 - P25$$

Example

For:

{1,2,3,3,5,6,7,9}

given:

$$Q1 = 2.5$$

$$Q3 = 6.5$$

Therefore:

$$\text{IQR} = 6.5 - 2.5$$

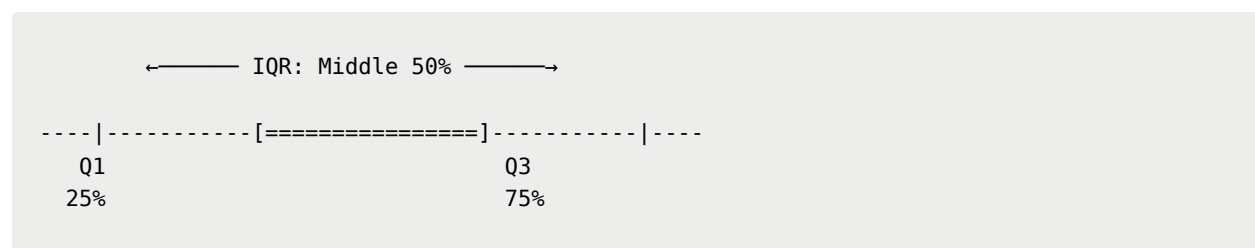
$$\text{IQR} = 4$$

Thus, the middle 50% of observations span **4 units**.

5. Usefulness in Identifying Spread

The IQR provides a **robust measure of variability** because it focuses only on the middle 50% of observations.

For example:



A **small IQR** indicates that the middle 50% of observations are relatively close together.

A **large IQR** indicates that the middle 50% are more widely spread.

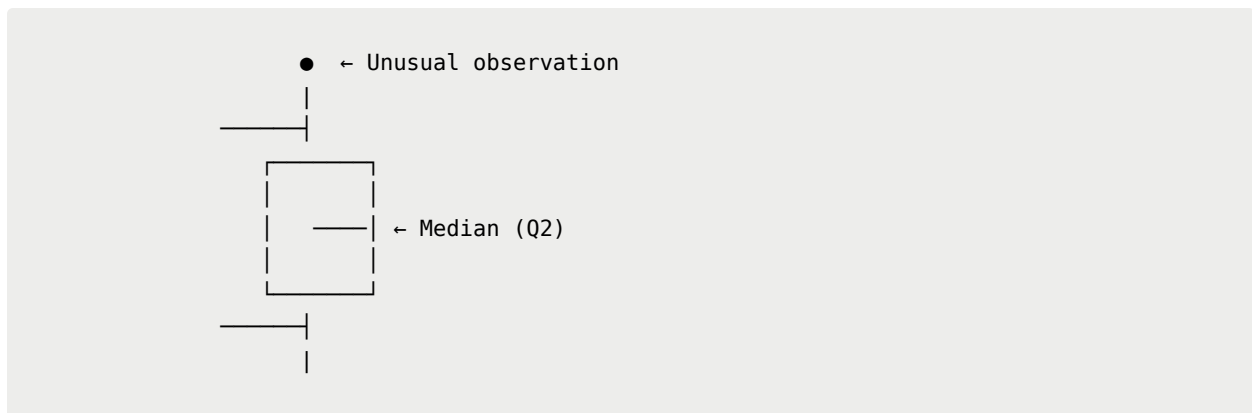
Unlike the range, the IQR is not strongly affected by observations at the extreme ends of the distribution. The notes specifically identify percentiles as a more **robust** approach to measuring variability than measures that are sensitive to outliers.

6. Identifying Unusual Observations

Percentiles and quartiles are especially useful for identifying **unusual or extreme observations**.

The **minimum and maximum values** can help identify potential outliers, but the range itself is highly sensitive to extreme values.

A **boxplot** uses quartiles and percentiles to provide a quick visual summary of the distribution:



Thus, quartiles and IQR form the basis for **boxplots**, which provide a quick way to visualize the distribution and highlight unusual observations.

7. Summary

Measure	Definition	Main Use
Percentile	Value dividing ordered observations according to percentage	Understand position within distribution
Q1	25th percentile	Lower boundary of middle 50%
Q2	50th percentile / Median	Central location
Q3	75th percentile	Upper boundary of middle 50%
IQR	$Q_3 - Q_1$	Measure spread of middle 50%

Key relationship

$Q_1 = P_{25}$, $Q_2 = P_{50}$, $Q_3 = P_{75}$

$IQR = Q_3 - Q_1$

Question 7:

Explain the concept of outliers in EDA. Discuss different methods for detecting outliers and explain how outliers affect statistical summaries.

Answer

1. Definition of Outlier

An **outlier** is a data value that is **very distant from the other values in a dataset**. The exact definition can be somewhat subjective, but several standard conventions are used in statistical summaries and plots.

An outlier does **not necessarily mean that the observation is incorrect**. It may represent a genuine extreme case, or it may result from errors such as incorrect units or faulty sensor readings. Therefore, outliers should normally be **identified and investigated** rather than automatically removed.

2. Methods for Detecting Outliers

A. Minimum and Maximum Values

The **minimum and maximum** values of a dataset can provide an initial indication of possible outliers.

For example:

10, 12, 13, 14, 15, 16, 100

Here, **100** is much larger than the other observations and should be investigated.

However, the **range** is very sensitive to extreme values, so it should not be used alone as a robust measure of spread.

B. Interquartile Range (IQR) Method

The **IQR** measures the spread of the middle 50% of observations:

$$\text{IQR} = Q3 - Q1$$

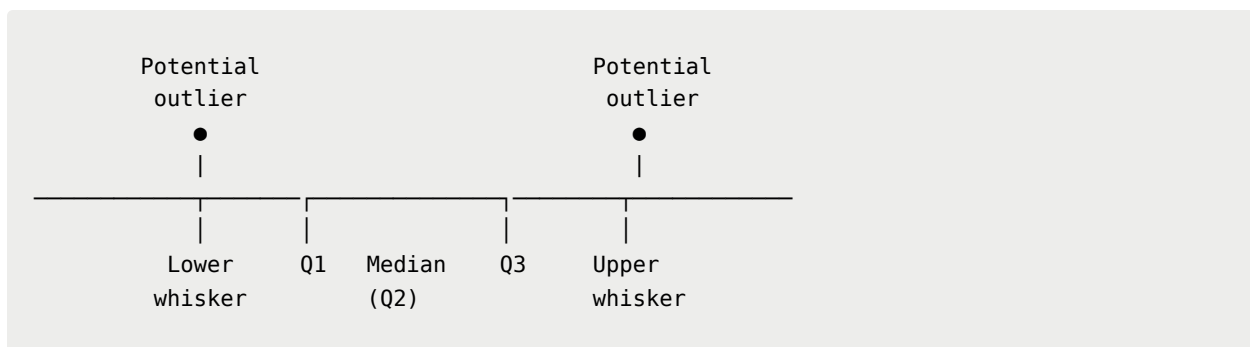
A commonly used boxplot convention identifies observations outside:

$$Q1 - 1.5(\text{IQR})$$

and

$$Q3 + 1.5(\text{IQR})$$

as potential outliers.



In standard boxplots, the whiskers extend to the furthest points within the **1.5 × IQR** limits, and observations beyond the whiskers are plotted individually and are often considered outliers.

C. Boxplot

A **boxplot** is a quick graphical method for detecting outliers.

It displays:

- **Q1** – 25th percentile
- **Median** – 50th percentile
- **Q3** – 75th percentile
- **IQR** = $Q3 - Q1$
- **Whiskers** – bulk of the data
- Individual points beyond the whiskers – potential **outliers**

Thus, boxplots make extreme observations easy to identify visually.

D. Histogram and Distribution Plots

A **histogram** provides a view of the overall distribution of numerical data. An unusually distant bar or a thin tail may indicate extreme observations.

A **density plot**, being a smoothed version of a histogram, can also help reveal unusual observations and the shape of the distribution.

E. Robust Statistical Measures

Another approach is to compare statistics that are sensitive to outliers with **robust statistics**.

For example:

- **Mean** → sensitive to extreme values.
- **Median** → robust to outliers.
- **Variance and standard deviation** → sensitive to outliers because they use squared deviations.
- **Median Absolute Deviation (MAD)** → robust to extreme values.

- **Percentiles/IQR** → relatively robust measures of variability.
-

3. Effect of Outliers on Statistical Summaries

Outliers can significantly distort statistical summaries.

A. Effect on Mean

The **mean** is strongly affected by extreme observations.

Example:

Without outlier:

10, 12, 13, 14, 15

$$\bar{x} = 645 = 12.8$$

With an outlier 100:

10, 12, 13, 14, 15, 100

$$\bar{x} = 1646 \approx 27.33$$

Thus, a single extreme value increases the mean from **12.8 to 27.33**.

B. Effect on Median

The **median** is much less affected by outliers.

For:

10, 12, 13, 14, 15, 100

the median is:

$$13 + 14 / 2 = 13.5$$

It remains close to the center of the main data.

Therefore, the median is called a **robust estimate of location** because it is not influenced strongly by extreme cases.

C. Effect on Variance and Standard Deviation

Variance and standard deviation are especially sensitive to outliers because they are based on **squared deviations** from the mean.

A very large deviation becomes even larger after squaring, causing the variance and standard deviation to increase substantially.

D. Effect on Range

The range is:

Range=Maximum–Minimum

Since it depends entirely on the extreme values, an outlier can make the range appear much larger than the typical spread of the data.

E. Effect on Robust Measures

Measures such as **median, IQR and MAD** are less affected by extreme observations.

Therefore, when outliers are present, robust measures can provide a more representative description of the typical data distribution.

4. Summary

Method/Measure	Effect/Use in Outlier Detection
Minimum & Maximum	Help identify extreme observations
IQR	Measures middle 50% and provides robust spread
Boxplot	Visually identifies observations beyond whiskers
Histogram	Shows unusual values and distribution tails
Density plot	Shows smooth distribution and extreme tails
Mean	Strongly affected by outliers
Median	Robust to outliers
Variance & SD	Highly sensitive because of squared deviations
Range	Extremely sensitive to extreme values
MAD	Robust measure of variability

Question 8:

Explain robust estimates of location and variability. Compare mean, median and trimmed mean with respect to their sensitivity to extreme values.

Answer

1. Introduction

In **Exploratory Data Analysis (EDA)**, estimates of **location** describe the central or typical value of data, while estimates of **variability** describe how much the data is spread out.

An estimate is called **robust** when it is **not sensitive to extreme values or outliers**. Robust estimates are useful when the data contains unusually large or small observations.

2. Robust Estimates of Location

The important estimates of location are **mean, median and trimmed mean**.

A. Mean

The **mean** is the sum of all values divided by the number of observations.

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

It uses **every observation**, including extreme values.

Therefore, the mean is **not robust** to outliers.

Example

Consider:

$$10, 12, 13, 14, 15$$

$$\bar{x} = \frac{10 + 12 + 13 + 14 + 15}{5} = 12.8$$

If an extreme value **100** is added:

$$\bar{x} = \frac{10 + 12 + 13 + 14 + 15 + 100}{6} = 27.33$$

Thus, the single extreme value changes the mean significantly.

B. Median

The **median** is the value such that approximately one-half of the data lies above and one-half lies below it. It is also the **50th percentile**.  BAD702-module-1-textbook(5)

For an ordered dataset:

$$x_1 \leq x_2 \leq \dots \leq x_n$$

the middle value gives the median.

For:

$$\text{Median} = \frac{13 + 14}{2} = 13.5$$

The extreme value 100 has very little effect on the median.

Therefore:

Median is a robust estimate of location

The notes specifically state that the median is **not influenced by outliers**.

 BAD702-module-1-textbook(5)

C. Trimmed Mean

A **trimmed mean** is calculated by **dropping a fixed number of the smallest and largest values** and then calculating the mean of the remaining observations.

 BAD702-module-1-textbook(5)

If p values are removed from each end:

$$\bar{x}_{trim} = \frac{\sum_{i=p+1}^{n-p} x_i}{n - 2p}$$

where the data is first sorted.

A common approach is to trim **10% from each end**.

The trimmed mean is a **compromise between the mean and median**: it is more resistant to extreme values than the mean, while still using more of the data than the median.

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3. Comparison of Mean, Median and Trimmed Mean

Property	Mean	Median	Trimmed Mean	
Uses all observations	Yes	No	No	
Effect of extreme values	High	Very low	Reduced	
Robustness	Not robust	Highly robust	Robust	
Calculation	Average all values	Middle value	Remove extremes, then average	
Information used	All data	Data position/rank	Most non-extreme data	
Best use	Data without significant outliers	Data with strong outliers	Data with some extreme values	

Sensitivity order

Mean > Trimmed Mean > Median

in terms of **sensitivity to extreme values**.

4. Robust Estimates of Variability

Variability measures describe the **spread of observations**.

Common measures include:

A. Interquartile Range (IQR)

The **IQR** is the difference between the 75th and 25th percentiles:

$$IQR = Q_3 - Q_1$$

It measures the spread of the **middle 50% of the data** and is relatively robust to outliers.

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B. Median Absolute Deviation (MAD)

The **Median Absolute Deviation from the median (MAD)** is a robust estimate of variability.

First calculate the median m , then calculate the absolute deviations:

$$|x_1 - m|, |x_2 - m|, \dots, |x_n - m|$$

The MAD is the median of these absolute deviations:

$$MAD = \text{Median}(|x_i - m|)$$

Like the median, **MAD is not influenced by extreme values.**

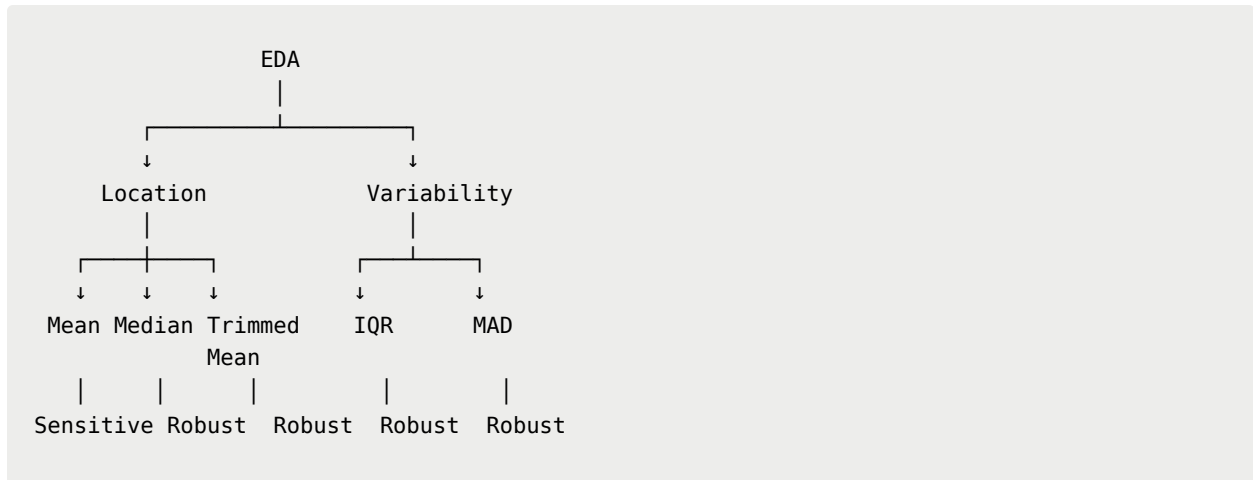
 BAD702-module-1-textbook(5)

5. Robust vs Non-Robust Variability

Measure	Sensitivity to Outliers
Range	Very high
Variance	High
Standard deviation	High
Mean absolute deviation	Not robust
IQR	Robust
MAD	Robust

The **variance and standard deviation** are especially sensitive to outliers because they are based on **squared deviations**.

6. Overall Concept



Question 9:

Explain correlation and covariance as measures for exploring relationships between two numerical variables. Discuss their interpretation with suitable examples/plots.

Answer

1. Introduction

In **Exploratory Data Analysis (EDA)**, **correlation** and **covariance** are used to study the **relationship between two numerical variables**.

If high values of X tend to occur with high values of Y, the variables are **positively associated**. If high values of X tend to occur with low values of Y, they are **negatively associated**.

2. Covariance

Covariance measures the extent to which two numerical variables **vary together**.

For variables X and Y , sample covariance is:

$$s_{XY} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

where:

- x_i, y_i = individual observations
- $\bar{x}, \bar{y}$ = sample means
- n = number of observations.

Interpretation

- **Positive covariance:** X and Y tend to increase or decrease together.
- **Negative covariance:** when X increases, Y tends to decrease.
- **Covariance near zero:** little linear co-movement.

However, the magnitude of covariance is **dependent on the units and scale of the variables**, so it is difficult to compare covariance values across different datasets.

3. Correlation

Correlation is a **standardized measure** of the association between two numerical variables. The most commonly used measure is **Pearson's correlation coefficient**. BAD702-module-1-textbook(5)

It is calculated as:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{(n - 1)s_x s_y}$$

where:

- s_x = standard deviation of X
- s_y = standard deviation of Y .

Correlation is essentially the standardized form of covariance.

Range

$$-1 \leq r \leq +1$$

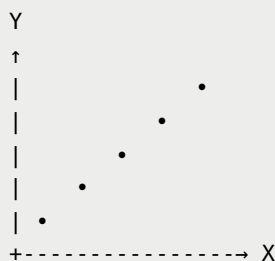
- $r = +1 \rightarrow$ perfect positive correlation
- $r = -1 \rightarrow$ perfect negative correlation
- $r = 0 \rightarrow$ no linear correlation

The correlation coefficient always lies between **-1 and +1**, making it easier to interpret than covariance.

4. Interpretation Using Scatterplots

The standard way to visualize the relationship between two numerical variables is a **scatterplot**. The **x-axis** represents one variable and the **y-axis** represents the other; each point represents one observation.

A. Positive Correlation

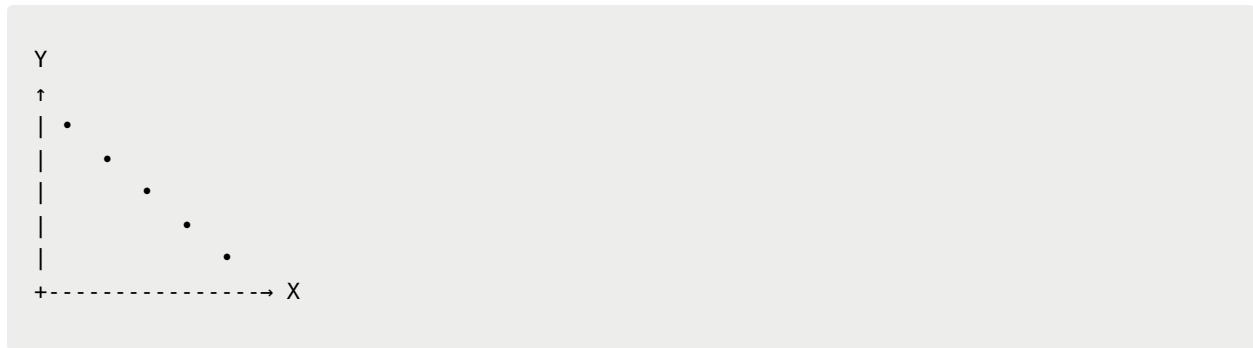


As X increases, Y generally increases.

$$r > 0$$

Example: **height and weight** generally show positive association.

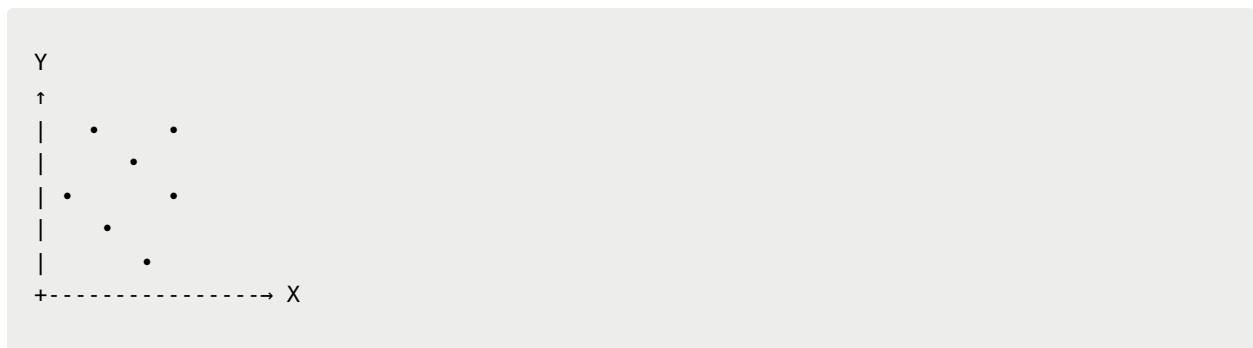
B. Negative Correlation



As X increases, Y generally decreases.

$$r < 0$$

C. No Linear Correlation



There is no clear **linear relationship**.

$$r \approx 0$$

An important limitation is that **correlation measures linear association**. A strong nonlinear relationship may have a correlation close to zero.

5. Correlation Visualization

For multiple numerical variables, correlations can be organized into a **correlation matrix**.

	X	Y	Z
X	1.000	0.80	0.20
Y	0.80	1.000	-0.40
Z	0.20	-0.40	1.000

The diagonal contains **1**, because each variable is perfectly correlated with itself. The matrix is symmetric, so the information above and below the diagonal is redundant.

6. Example

Consider:

$X = \{1, 2, 3\}$

$Y = \{4, 5, 6\}$

As X increases, Y also increases perfectly.

Therefore:

$r = +1$

This represents a **perfect positive linear correlation**.

For a real-world example, the notes discuss the daily returns of **AT&T** and **Verizon**. Their scatterplot shows a positive relationship: on many days both stocks tend to move in the same direction.

7. Correlation vs Covariance

Feature	Covariance	Correlation
Meaning	Measures how two variables vary together	Measures strength and direction of association
Range	No fixed range	-1 to +1
Units	Product of units of XX and YY	Unitless
Standardized	No	Yes
Interpretation	Depends on scale	Easier to interpret
Visualization	Often examined using scatterplot	Scatterplot/correlation matrix

8. Important Limitation

Correlation does not imply causation.

Also, Pearson correlation may not adequately describe a **nonlinear relationship**. Therefore, the correlation coefficient should be interpreted together with a **scatterplot** rather than used alone.